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M 005 CONTROL VALVES

1 SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2 Scope of Work ; including but not limited to:

- Control Valves (electric motor and pneumatic operated) including its associated electrical equipment as shown, on the Flow Diagrams and others as required by the Bidder to make the systems complete and functional.
- The valves together with all appurtenances and accessories as described in the Control Valve Data Sheets and as herein after specified or required by the Bidder's design

Bidder shall furnish all special erection tools and instruments required for erecting and maintaining the equipment.

2 CODES & STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 'Codes and Standards'. The Codes and Standards of the following organizations and other international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this Specification.

ASTM	American Society of Testing and Materials
ASME	American Society of Mechanical Engineers
NFPA	National Fire Protection Association
ANSI	American National Standards Institute, Inc.
FCI	Fluid Controls Institute
IEC	Electro Technical Commission
IEEE.	Institute of Electrical and Electronic Engineers
MSS	Manufacturers Standardization Society of the Valve and Fitting Industry

SAMA	Scientific Apparatus Manufacturer Association
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3 DESIGN AND OPERATING CONDITIONS

All control valves shall be capable of operating without cavitation in the control range indicated in the control valve data sheet included in the Exhibit 1 to be submitted by the Bidder.

Valves shall be a high lift design capable of handling the maximum flow requirements set forth in the control valve data sheets at 70 to 80 percent of the total stem travel.

No control valves shall be released for fabrication until the Bidder has submitted the control valve data sheet included as Exhibit 1 herein and this has been approved by the Owner.

For environmental considerations see Part 4, Subsection 2.16.

4 DETAILED REQUIREMENTS

4.1 General

Control valves shall be furnished in accordance with the requirements of this specification and the applicable requirements of Technical Specification M-004 Valves.

Control valves shall be pneumatic or electric actuated unless otherwise noted or required to meet the operating conditions. Each actuator assembly shall have ample capacity to position the valve to meet all operating conditions set forth in the Specification or as required by Bidder's design with the valve and actuator oriented in any direction within the piping system.

Where the 0.2 to 1.0 bar (g) nominal control air pressure is transmitted directly to the diaphragm of the Bidder's control valves, the sizing of the actuator spring shall correspond to the nominal control air pressure range. Where positioners are used the diaphragm air pressure shall be 0.2 to 1.0 bar (g) or 0.4 to 2.0 bar (g) depending on the application. Bidder's control valves shall be capable of successful operation in the applicable range. Springs shall be enclosed and shall include provisions for external adjustment of the spring compression.

Valves shall be of the flow to open or balanced type, unless otherwise specified, with actuator area and spring design sufficient to close against maximum pressure differential encountered when valves are closed. Sufficient seat loading force shall be provided on single-ported valves to prevent wire drawing damage to the valve seats.

Valves shall be single-ported whenever possible.

Nitriding treatment or chrome plating on any surface exposed to the process fluid is prohibited.

Valve ports shall be designed to meet the specified service conditions in each case and shall be restricted in area where necessary to provide adequate regulation under all loads. Valve port restrictions shall be accomplished by reduction in actual flow area in order to provide lowest minimum flow.

Valve body size for modulating-type control valves may be smaller than the line size under the following conditions:

Plug guiding shall be sufficiently rugged to withstand the increased inlet velocity.

Body size in no case shall be more than two pipe sizes smaller than the line size. Valves having 90 mm and 125 mm nominal body sizes will not be considered.

Allowable pressure drop shall not be exceeded. When allowable pressure drop is exceeded, the Bidder shall provide cavitation resistance design and trim.

Valve size for on-off service shall be full line size.

The valve pressure-temperature ratings shall be per ANSI B16.34.

Valves and operators shall function smoothly, without sticking, rubbing, scoring or vibrating on opening or closing or at any valve position.

Each pneumatic operated valve shall be furnished with an air filter/pressure regulator set with air-pressure indicator and a relief valve, except for self-operated valves.

It is the intent that the valves shall fail safe, either open or closed, as Bidder shall indicate in the control valve data sheets, in event of failure of

air signal from any valve component (pressure controls, positioner, etc.), failure of the main control air supply, or in case of power failure.

The control signal from the applicable I/P converter/transmitter shall be transmitted to Bidder's control valves in the range specified. Bidder's control valves shall be capable of successful operation in this range.

4.1.1 Mounting

The valves shall be suitable for mounting in horizontal, vertical or inclined run of pipe, with the stem in any position required to suit the pipe arrangement

Welded-in seats, if furnished, shall be designed for field reseating. Special tools for reseating welded-seat valves shall be provided by the Bidder.

Positioners, pressure controllers, limit switches and other valve appurtenances shall be shipped mounted on the valve. Temperature controllers shall each be tagged with the Owner's tag number and shipped unmounted. I/P converters/transmitters shall not be mounted on.

Valve size shall be designated by the actual body size.

Control valves shall be furnished with positioners where:

- Small signal changes are to be amplified.
- Split range control is required.
- Valve pressure drop is excessive.

Positioners shall have a feedback element to characterize relationship between input signal and final element position.

4.2 Valve Trim

Trim for all valves shall be Type 316 stainless steel or other suitable material.

Plugs for globe and angle valves shall be top and bottom-guided, top-guided or cage-guided. The most rugged type of plug guiding shall be provided for the specified operating conditions.

Guiding in other valves, such as ball valves, shall be a proven type.

Each top-guided valve shall have a heavily constructed plug that fits snugly in an elongated bonnet guide bushing to insure perfect disc-to-seat alignment under the most severe operating conditions.

Valve plugs and seats or cages shall be contoured for the flow control characteristic specified or to suit special applications. Characterizing positioner cams may be used when necessary.

The valve plug shall be solidly connected to its stem in each case. The threaded and pinned type is acceptable, however, no threads shall be exposed.

Seat rings shall be replaceable. Seat rings for cage-guided trim shall be securely locked in place to prevent leakage past the body-seat ring joint. Screwed-in seat rings shall be provided for other types of guiding. Screwed-in seat rings shall be seal welded for applications rated Class 600 and higher or where temperature cycling can be expected.

Where required for the service conditions, the seat rings and the seating surface of the inner valves shall be faced with Stellite or other acceptable hardened steel.

4.3 Body, Bonnet and Packing.

In addition to the applicable requirements of M-004 Technical Specification, Valves, the following shall apply. In case of conflict the more stringent requirements shall apply.

4.3.1 Cast Iron Valves

Cast iron valves in sizes 50 mm and larger shall conform to the requirements of:

MSS SP-70 - Cast Iron Valves, Flanged and Threaded Ends.

MSS SP-85 - Cast Iron Globe and Angle Valves, Flanged and Threaded Ends.

Other types and sizes of cast iron valves shall also conform to the applicable requirements of the above MSS Standards and shall be acceptable to the Owner. Pressure-temperature ratings of all cast iron valves shall be in conformance with those set forth in Table 1 of MSS SP-70 and MSS SP-85, unless otherwise indicated.

4.3.2 Steel Valves

Pressure-Temperature Ratings: Valves shall have pressure- temperature ratings equal to or greater than that in ANSI B16.34, except the ratings of the special Class shall not be used for socket-welded or threaded end valves.

Valves shall be designed and manufactured in accordance with the following:

Butt-Welding and Flanged End Valves: ANSI B16.34.

Socket-Weld or Threaded End Valves: MSS SP-84.

Special design valves for pressure-temperature conditions exceeding the allowable ratings or Classes set forth in ANSI B16.34 shall be permanently marked on the valve body with the specified pressure-temperature rating for which Bidder shall certify that the valve is designed and, in each such case, that the valve conforms to the applicable requirements of the governing code(s).

4.3.3 Bronze Valves Shall Conform To MSS SP-80.

Valves shall be provided with Teflon or Teflon-asbestos packing for service conditions not exceeding 232°C. Packing for service conditions in excess of 232°C shall be designed for the service and acceptable to the Owner. If Teflon packing is used for vacuum service, it shall be the double-reverse type, suitable for both vacuum and pressure.

All valves shall be provided with extended bonnets and packing where required for the maximum specified or required temperature conditions.

Each valve shall be provided with a valve stem travel and position indicator.

Globe and angle valve designs in each case shall permit the valve to be installed and to function properly in any position in the pipe run (horizontal, vertical, inclined, stem up, down, side etc.). Valve position restrictions, if any, are regarded as essential Bid Proposal and design information and must be included as part of the Bid Proposal and thereafter on the valve drawings submitted for the Contract.

4.4 Weld Ends, Extensions and Reducers

Alloy steel valves that are to be welded to carbon steel piping shall be provided with 150 mm long carbon steel pipe nipples or reducers, as applicable, welded to valve inlets and/or outlets in the shop.

Welding of valve end attachments and weld examinations shall be in accordance with the applicable requirements of ANSI/ASME B31.1.

The type of welding shall be as specified in the applicable standard as accepted by the Owner. Welding procedures shall be submitted to the Owner for review.

Face-to-face dimensions of flanged valves and end-to-end dimensions of butt-welding valves shall conform to the applicable requirements of ANSI B16.10.

Butt-welding end preparation shall conform to the applicable details given in this specification and/or as accepted by the Owner. Where special wall pipe is indicated for butt-weld end valve connections, the Bidder shall provide appropriate weld end details with the actual dimensions of the pipe and weld end preparation for Owner approval prior to machining of the ends of such valves.

Socket-welding end preparation shall conform to the applicable requirements of ANSI B16.11. Where special wall pipe is indicated for socket-weld valve connections, the following shall apply.

Hubs of socket-weld ends shall be machined to suit the O.D. of the special pipe wall, with clearance and tolerance comparable to that set forth in ANSI B16.11. for schedule pipe connections.

Bore of socket-weld ends shall be equal to or greater than the O.D. of the special wall pipe, with tolerances comparable to that set forth in ANSI B16.11. for schedule pipe connections, unless otherwise accepted by the Owner.

The wall thickness of socket weld ends (including hubs), after machining, shall not be less than the allowable minimum set forth for fittings in Article 6 of ANSI B16.11.

Where the body size of the valve offered differs from the pipe size indicated in Item 23 of the Control Valve Data Sheet (prepared by Bidder) or where the inlet and outlet pipe sizes are different, any required reducer(s) shall be furnished by the Bidder and shall be welded to the valve and heat treated and stress-relieved, as required, in the factory. All nipples provided shall be heat treated and stress-relieved in the factory.

4.5 Temperature Controllers

Temperature controllers shall be of the indicating type. Temperature controllers shall normally be provided in the plant DCS whenever practicable.

Controllers shall be of the adjustable proportional type with adjustable reset and rate as required to suit the service. Set point shall be adjustable and indicated in the engineering units.

Deviations, from the standard drawings for the use of the manufacturer's standard wells shall be made only with approval of the Owner. Well mounting threads shall be as specified on the standard drawings.

When the control temperature set point is within 55°C of ambient, a Class 1A (SAMA Standard) liquid-filled thermal system shall be used instead of the 'vapor-filled type specified

Controllers shall be suitable for surface mounting or for yoke mounting on the valve, as required.

Temperature controllers shall be furnished with a temperature sensing device having capillary tubing no more than 9.2 m long, and with stainless steel temperature wells. The capillary tubing shall be furnished with a spiral armor casing for protection. Each temperature controller shall be of the pneumatic pilot proportional band plus reset type with integral temperature indication except as otherwise noted. The temperature controllers, where required, shall be specified on the Control Valve Data sheets.

4.6 Level Controllers

Level controllers shall be furnished in accordance with the requirements of, Technical Specification I-001 Control and Instrumentation Equipment.

4.7 Pressure Controllers

Pressure controllers shall be in the plant DCS whenever practicable. All local pneumatic pressure controllers shall be provided with bronze or stainless steel Bourdon tube or bellows as required by the pressure to be controlled. Each controller shall be the pneumatic pilot proportional plus reset type, with adjustable proportional band and with adjustable reset and with integral pressure indication except as otherwise noted. Pressure controllers, where required, shall be indicated on the Control Valve Data Sheets.

Pressure controllers shall be provided with pressure elements having adequate sensitivity for required control pressure plus the ability to withstand the maximum pressure to which the pressure element may be subjected under all conditions. This can be accomplished by the use of a heavy pressure element, an overpressure protection device, or a combination of each. Process conditions, measuring range and set point data are to be submitted by the Bidder.

Each controller shall be of the pneumatic pilot proportional plus reset type with adjustable proportional band and with adjustable reset. Reset adjustment shall include provisions to cut out reset during startup. Each controller shall include control point adjustment with calibrated pressure setting dial and air pressure gauges indicating “supply” and “diaphragm” pressures.

Controllers shall be suitable for surface mounting or for yoke mounting on the diaphragm valve, as required.

4.8 Control Valve for Bypass System

Each control valve shall be sized and selected to provide reliable operation and control at the specified operating and design conditions. Valve body shall be designed to withstand thermal cycling. Thin walled sections machined inside and outside shall be provided to reduce thermal fatigue. Transitions between thick and thin wall sections shall be gradual. The operating and design conditions are specified on the attached Control Valve Data Sheets.

Control valve shall not have any internal bolts to lock the seat to the body or the outlet diffuser.

The inlet cage shall be an integral part of the Bonnet and shall not rest on the seat.

Pressure seal bonnets shall be offered for ease of maintenance and better seating between body and bonnet.

The seat shall not be used as a pressure reducing point during low lifts of the plug.

The valve design shall be as to avoid any internal gasket, thereby eliminating any leakage point through the valve body.

For Pneumatic actuators only piston type shall be used. Diaphragm actuators are unacceptable.

The valve actuator yoke design shall be such that it will accept installation of all required accessories, such as air sets, limit switches, and position indicators.

The plant air supply pressure shall be per attached Control Valve Data Sheets.

Actuators shall be sized for stroking the valve against 120% of the maximum differential pressure.

Handwheel along with Limit Switch (Open & Close Position), AFR, stainless steel tubing shall be offered.

Positioners shall be of the Electro-pneumatic type and shall be provided with pressure gauges for indicating the positioner outlet and control signal pressures.

Positioners shall be designed for a control signal input range as specified on the valve data sheets.

All positioners shall be equipped with filter regulator air sets and shall be mounted on the valve yokes. If the positioner can accept the maximum air supply pressure, then the positioners shall be equipped with an air filter in place of a filter regulator.

The positioner action shall be field changeable without additional parts.

The positioners shall be of the force balance type, employing positive cam type feedback to provide accurate control positioning.

All other accessories such as handwheels, limit switches, solenoid valves etc. shall be in accordance with the specifications in the attached valve data sheet. The actuator shall be able to open and close in < 10 sec in both direction through the positioner.

The material and design of the seat, trim and stem shall be sufficient to resist wear under severe service conditions. Cage material shall be of a high alloy material X19CrMoVnBn 111 for good thermal cycling and erosion resistance. Cylindrical cages shall not be provided.

The plug offered shall have holes drilled at the bottom portion of the plug (10%) to allow for steam to be controlled at low lifts.

The use of stellite overlayed over 316 SST is not acceptable

The seat shall be seal welded to the body to avoid any leakage point between the valve body and the seat. The welding shall not be directly between the body and the seat, instead a dual seat design shall be used to take care of thermal compensation and eliminate seat breakage and leakage through the valve.

Pressure reducing cages/pipes shall be provided below the seat. These pressure reducing cages shall not be welded on both side to allow for thermal transient. The bottom of the pressure reducing cages shall be blocked and should not allow any steam flow through the bottom to ensure excellent mixing of steam and water droplets.

Spray water shall be admitted at the outlet of the valve using Steam assist spring loaded nozzles mounted around the circumference of the valve outlet. No diffuser plates shall be allowed downstream of the water injection.

The nozzle design shall utilize inlet steam to inject along with the cold water. The angle of the nozzle seat should be slightly different from the nozzle plug to increase the water velocity while injecting the water into the system. Separate actuated control valve with Solenoid valve shall be provided to control the inlet steam flowing into the steam assist spring loaded nozzles.

The stem and the plug shall be connected using the shrink fit method. Threaded connection, cotter - pin arrangement are not acceptable.

Valve packing shall be selected for the design conditions of the service intended. Teflon packing shall not be used in applications with operating temperature of the application above 450 F. Packing box design shall consist of two independent sets of rings separated by a metallic spacer. Control valves shall be sized for 110% of a maximum flow rate or 130% of the normal operating flow rate, which ever is greater

Control valves shall be capable to operate within the specified flow range between 10% and 90% of travel.

Noise level within 3 feet of the valve shall not exceed 85 dBA

Valve stroke length shall be 90mm or higher for better resolution.

4.9 Valve Limit Switches

All control valves shall have limit switches. One limit switch shall close when the valve is in the full open position and one limit switch which shall close when the valve is in the fully closed position, this shall be indicated on the Control Valve Data Sheets. The limit switches shall be in accordance with Part 4, Technical Specification E-210, Low Voltage Motors.

4.10 Handwheels

Handwheels shall be provided with all control valves. These shall be capable of opening and closing the valves with a handwheels rim pull of less than 36 kg when the valve is experiencing the maximum shutoff conditions specified in the data sheets.

4.11 Valve Accessories

4.11.1 Positioners and Booster Relays

Valve positioners and booster relays shall be provided for all modulating valves where pressure, temperature or unbalance are excessive, where specifically called for in the Specification, or where satisfactory operation of Bidder's hardware and/or systems can only be ensured by their use. Bidder shall be responsible for discussing the systems with the Owner and discussing the necessary engineering data required to determine the need for positioners or booster relays.

Bidder shall bear the costs for additional positioners and relays required to obtain satisfactory hardware and system operation at initial installation.

Positioner shall be mounted on the valves and receive mechanical position feedback from valve stem travel.

Positioner shall include filter regulator to reduce the instrument air system header pressure to the pressure required by the positioner output pressure range.

Positioners shall be contained in IP 66 enclosures.

Positioners for local control loops shall be pneumatic and receive 0.2 to 1 bar (g) or 0.4 to 2 bar (g) control signal.

Electro pneumatic positioners shall be provided for valves controlled by a 4 to 20 mA dc signal, where required by the Bidder's control system design.

Positioner bypass manifolds shall be furnished except where the instrument signal cannot fully stroke the valve or where a split range signal is utilized.

4.11.2 Hydraulic Snubbers

Hydraulic snubbers, where required or where indicated in this specification, shall be provided. The needle valve of the snubber shall be adjustable and locking type to provide various damping effects.

4.11.3 Electro Pneumatic Converters

Electro-pneumatic converters shall be provided where required by the Bidder's control system design. Range and zero adjustments shall be provided, and the device shall operate using a 4~20 mA dc signal. The converter shall be mounted on the control valve unless prohibited by valve configuration or operating conditions (temperature, vibration etc.).

4.11.4 Pneumatic Supply Pressure Regulators

Supply pressure regulators shall be as specified herein. Common pressure regulators may be used where suitable for the accessories mounted on the valves. All accessories that are required to be mounted on the valves shall be shop mounted and properly adjusted prior to shipment. All air supply and signal tubing provided integral with the control valves, controllers, positioners,

etc., shall be copper or stainless steel tubing of not less than 0.9 mm wall thickness. Plastic tubing is not acceptable.

Each supply pressure regulator shall consist of a combination filter regulator with an integral relief valve and drip well with drain cock.

The reduced pressure range shall be adjustable and as required for the application.

Connections shall be 8 mm. NPT.

Integrally mounted regulator outlet pressure gauges shall be included unless the outlet pressure is indicated by gauges on equipment using gas from the regulator.

4.11.5 Solenoid Valves

Enclosures for solenoid valves shall be IPW66 (NEMA 4) for outdoor locations. Solenoid valves shall be rated for continuous duty at 110 V ac or as specified otherwise. Solenoid coils shall have Class H insulation. Solenoid valves shall be specified on the Control Valve Data Sheets. All solenoid valves shall also be provided with manual operators. The valves shall be 3-way with minimum 8 mm NPT connection.

Solenoid valves shall have body construction, trim materials and internal arrangements designed for the application.

For instrument air, solenoid valves shall have brass bodies, composition seats and universal flow design.

4.11.6 Position Transmitters

Bidder shall provide position transmitters with the control valves as required. The transmitters shall provide a 4~20 mA signal for Bidder use.

4.12 Valve Sound Levels

4.12.1 General

The valves shall meet the noise criteria specified in Part 4, Subsection 2.9 and the following:

All sound levels given in this Specification refer to levels measured one meter downstream of the valve discharge plane and one meter away from the exterior wall of the discharge piping, and in accordance with ANSI Standard.

Noise reduction effects of thermal insulation are not to be considered.

Bidder shall state in his Bid Proposal the guaranteed maximum noise level of each valve over its operating range.

Bidder shall specify his method for determining sound levels as well as specifying the operating parameters used in calculating the valve sound levels (e.g. fluid temperatures and pressures, pressure drops, pipe wall thickness, flow rates etc.).

4.13 Cleaning and Painting

Refer to M-004, clause 4.12.

4.14 Protection

The protection of the valves shall be in accordance with Specification M-004, Clause 4.13.

4.15 Valve List

The Valve List shall be prepared by the Bidder in accordance with the requirements of Technical Specification M-004 Clause 4.15.

4.16 Control Valve Data Sheet

The Control Valve Data Sheet shall be prepared by the Bidder in accordance with the requirements of Exhibit 1 of this Technical Specification.

Data sheets for control valves shall be prepared by the Bidder as per Exhibit 1 herein.

5 PERFORMANCE TESTING AND GUARANTEES

After equipment is put in service the Bidder shall run a performance test on the equipment to certify that it performs according to the guarantee in every respect. Any deficiencies brought out by this test shall be promptly corrected by Bidder at no additional cost to Owner.

5.1 Hydrostatic Shop Tests

All control valves shall be hydrostatically and seat tightness tested at the manufacturer's plant.

Pressure testing of the shell and seat (closure) of each steel valve shall be performed in accordance with the requirements set forth in ANSI except gas seat B16.34 (closure) test if not acceptable in-lieu of hydrostatic test and duration time and maximum permitted leakage rate for seat (closure) tests shall be as forth in MSS-SP-61.

Pressure testing of the shell and seat of each cast iron valve shall be performed in accordance with the applicable requirements set forth in MSS-SP-70, MSS SP-7 and MSS-SP-85.

In general valves with bypasses shall be tested with bypasses in place. However, in cases where shop hydrostatic valve body test pressures exceed allowable working pressure for bypass piping the hydrostatic tests shall be made without bypass in place. After each valve is installed, the entire piping system shall be given a field hydrostatic pressure tests at 1.5 times the design pressure at ambient water temperature, and valve bypass shall withstand this test. Any defects or leaks detected in the bypass during this test shall be remedied by Bidder at no additional cost to Owner.

Testing other than specified herein shall comply with the requirements as mentioned in M-014 inspection and test.

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control for all control valves shall be as per the general technical requirement Part 4 Subsection 2.7.

7 INSTRUMENTATION AND CONTROLS

Additional Instrumentation and Control requirements shall be as specified in Part 4, Technical Specification, I-001.

8 ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for control valves. These equipments shall include but not limited to the following:

1. Motors.
2. All electrical control, monitoring and instrumentation devices, and other appurtenances required for this Specification, M-005 Control Valves, shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 Low Voltage Motors.

EXHIBIT 1 CONTROL VALVE DATA SHEET

SERVICE	1. SPECIFICATION NUMBER :		REL NO:		6. TAG NO. & VALVE DESCRIPTION																						
	2. PIPE CLASS:		SEISMIC CLASS:																								
	3. P & ID :																										
	4. PIPE DESIGN TABLE																										
	5. DATA TABLE :		NONE																								
	7. FLUID:		RAW WATER		OIL NO		STEAM																				
			DEMIN WATER		AIR		NATURAL GAS																				
	8. FLOW :		LB/HR	GPM	SCFM	MAXIMUM FLOW	NORMAL FLOW																				
	9. TEMPERATURE		(°C)																								
	10. INLET PRESSURE bar(g)																										
	11. OUTLET PRESSURE bar(g)																										
	12. SPECIFIC GRAVITY :																										
	13. 40 YEAR RADIATION EXPLOSURE (PLUS ACCIDENT IF SAFETY-RELATED) : Not Applicable																										
	14. SERVICE CONDITION :		MODULATING		ON-OFF																						
	15. MAX. SHUTOFF P (bar)MAX TEMP . (°C).....OCCURS				SIMULTANEOUSLY																						
	16. DESIGN PRESSURE bar(g)		DESIGN TEMPERATURE (°C).....																								
	17. DOWNSTREAM PIPE INSULATION THICKNESS :		mm																								
BODY	18. MINIMUM BODY SIZE																										
	19. BODY TYPE		GLOBE		OTHER		DOUBLE – PORTED VALVE PRESSURE CLASS (ANSI)																				
	20. BODY MATERIAL :		PER PIPE DESIGN TABLE		OTHER																						
	21. BODY INSULATION THK		INLET		OUTLET																						
	22. TYPE AND CONECTION		BUTT WELD		FLNGD		BUTT WELD																				
			SOC. WELD		SCRWD		SOC. WELD																				
			FLANGELESS				FLANGELESS																				
	23. CONNECTING PIPE SIZE / SCHEDULE																										
	24. WELD END STANDARD (BUT WELD ONLY)		OTHER		OTHER																						
	25. REDUCERS FROM VALVE TO CONNECTING PIPE SIZE REQUIRED		YES		NO		YES																				
			MATERIAL		MATERIAL																						
	26. CHARACTERISTICS:		LINEAR		PERCENT		QUICK OPENING																				
	27. STUFFING BOX PACKING :		SINGLE		DOUBLE																						
OP.	28. AIR FAILS, VALVE		OPEN		CLOSES		29. ELEC. POWER FAILS, VALVE :																				
							OPEN																				
							CLOSES																				
	30. MAXIMUM STROKING TIME (SEC) : OPEN		CLOSE		BY VENDOR																						
ACCESS	31. POSITIONER :		YES		NO		BY VENDOR																				
	32. CONTROLLER DATA SHEET NUMBER :																										
	33. CONTROLLER LOCATION :		VALVE YOKE		REMOTE																						
	34. LIMIT SWITCHES :		NONE		FULL OPEN		FULL CLOSED																				
							%FULL OPEN																				
	35. SOLENOID VALVE		NONE		AC		DC																				
					SPECIAL (SEE MARK)		%FULL OPEN																				
	36. SOLENOID VALVE LOCATION		VALVE YOKE		REMOTE																						
	37. E/P CONVERTES:		NONE		INPUT SIGNALTO.....																						
	38. E/P CONVERTES LOCATION :		REMOTE		VALVE YOKE																						
39. POSITION TRANSMITTER :		NONE		YES (SEE REMARKS)																							
40. HAND WHEEL:		NONE		TOP-MOUNTED		SIDE-MOUNTED																					
41. AIR SET (W/GAGES):		REQUIRED FOR EACH		AIR SUPPLY		RANGE AsRequired.....																					
42. SPECIAL ACCESSORIES:																											
REMARKS																											
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